PREOPERATIVE DIAGNOSIS: Age-related nuclear sclerotic cataract (grade 3+), visually significant, right eye.

POSTOPERATIVE DIAGNOSIS: Nuclear sclerotic cataract, right eye.

PROCEDURE: Phacoemulsification cataract extraction with posterior chamber intraocular lens (PCIOL) implantation, right eye.

PROCEDURAL TECHNIQUE: The patient was brought to the ophthalmic operating suite, positioned supine, and monitored. Topical proparacaine 0.5% and 5% betadine were instilled into the right eye, which was draped in sterile ophthalmic fashion with an eyelid speculum positioned.

Under high-magnification operating microscope visualization, a 1.0 mm paracentesis was created at the 2 o'clock limbus. Intracameral 1% unpreserved preservative-free lidocaine and 0.1 mL of 0.025% trypan blue were injected to stain the anterior capsule. Viscoelastic (Healon GV) was instilled to maintain anterior chamber depth. A 2.4 mm clear corneal temporal incision was constructed using a diamond blade. A continuous curvilinear capsulorhexis (CCC) measuring 5.2 mm was completed using Utrata forceps.

Hydrodissection and hydrodelineation were performed using balanced salt solution (BSS) on a Chang cannula, confirming free rotation of the lens nucleus. Using the Centurion phacoemulsification system, the nucleus was divided into four quadrants using a stop-and-chop technique. The nuclear fragments were emulsified and aspirated using torsional ultrasound with vacuum of 450 mmHg and ultrasound power of 35%. Cortical clean-up was executed with bimanual irrigation/aspiration handpieces until the capsular bag was completely clear.

The capsular bag was reinflated with cohesive viscoelastic. An Alcon AcrySof IQ monofocal hydrophobic acrylic intraocular lens (+21.5 diopters, target -0.25 D) was loaded into an injector cartridge and smoothly unfolded in the capsular bag with haptics oriented at the 3 and 9 o'clock meridian. Viscoelastic was thoroughly aspirated from behind and in front of the lens. The corneal incisions were hydrated with BSS to achieve watertight closure; Seidel testing was negative. Subconjunctival cefuroxime and dexamethasone were instilled.